

Manish Patel

PhD Candidate | Active Matter & Non-equilibrium Statistical Physics

Institute of Physics, Bhubaneswar, India | Homi Bhabha National Institute, Mumbai, India

Expected PhD completion: January 2027

Date of Birth: 19 July 1997

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Google Scholar

ResearchGate

Webpage

Academic Profile

PhD in Physics

Institute of Physics, Bhubaneswar & Homi Bhabha National Institute, Mumbai

📅 Expected January 2027

📍 Bhubaneswar, India

Supervisor: Prof. Debasish Chaudhuri

Diploma in Advanced Physics

Institute of Physics, Bhubaneswar

📅 2021 - 2022

📍 Bhubaneswar, India

Percentage: 80.91% – **Certificate**

Project *Stochastic Dynamics and Thermodynamics of Active Brownian Particles*

Supervisor: Prof. Debasish Chaudhuri

M.Sc. Physics

University of Delhi

📅 2017-2019

📍 New Delhi, India

Percentage: 65.05% – **Certificate**

Thesis: *Redshift Evolution of Quasars:* **Supervisor:** Prof. T. R. Seshadri

B.Sc. Physics (Hons.)

Hansraj College, University of Delhi

📅 2014 - 2017

📍 New Delhi, India

Percentage: 89.0% – **Certificate**

Project: *Water Treatment Using Light-Activated Nanoparticles:* **Supervisor:** Dr. Maya Verma

Research Interests

Active Matter

Inertial Dynamics

Stochastic Thermodynamics

Interacting Active Systems

Biological physics

Nonreciprocal Dynamics

active bath

Nonequilibrium Statistical Physics

Research Experience

I am a PhD candidate in the group of Prof. Debasish Chaudhuri at the Institute of Physics, Bhubaneswar. My research lies at the intersection of nonequilibrium statistical physics, active matter, and biological physics, with a focus on the stochastic dynamics and thermodynamics of active systems.

A central theme of my research is to understand the physical mechanisms underlying transport, fluctuations, and dynamical transitions in systems far from equilibrium. I am particularly interested in the interplay of activity, inertia, confinement, interactions, and competing dynamical timescales, and in how these ingredients give rise to phenomena that have no direct equilibrium analogue. I combine analytical theory, stochastic dynamics, and physically motivated modeling to develop quantitative descriptions of such complex nonequilibrium behavior.

My work has focused on developing analytical approaches to characterize non-Gaussian fluctuations, re-

entrant phenomena, anomalous and non-monotonic diffusion, and crossovers between distinct dynamical regimes. I am also interested in how interactions and non-reciprocal couplings modify the dynamics and transport properties of active systems. These questions provide a natural connection to biological physics, where persistent motion, fluctuations, non-reciprocal interactions, and nonequilibrium driving are intrinsic features of living systems.

My experience with analytically challenging stochastic systems and multiple competing timescales motivates me to explore increasingly complex nonequilibrium phenomena, particularly those arising in living matter. I am interested in understanding how microscopic fluctuations, interactions, and active processes become coupled across multiple spatial and temporal scales, and how such coupling gives rise to collective behavior and emergent dynamics. I view the inherent complexity of biological systems as an exciting theoretical challenge, where rigorous statistical-physics approaches can help identify the mechanisms underlying collective dynamics and function.

My broader research goal is to develop a quantitative understanding of complex nonequilibrium systems, combining analytical rigor with physically informed modeling and, where possible, connections to experimentally accessible observables. I aim to build on my background in active-matter dynamics and nonequilibrium statistical physics to address challenging problems in active matter, soft matter, and biological physics, particularly those involving coupled processes across scales, strong fluctuations, and intrinsically nonequilibrium dynamics.

Publications

1. *Non-monotonic diffusion from nonequilibrium driving*
Manish Patel, Ritwick Sarkar, Urna Basu, Debasish Chaudhuri
[arXiv:2607.25902](#) (2026).
2. *Crossover dynamics and non-Gaussian fluctuations in inertial active chains*
Manish Patel, Subhajit Paul, Debasish Chaudhuri
J. Chem. Phys. **164**, 134110 (2026).
3. *Controlling inertial active Brownian motion via stochastic resetting*
Manish Patel, Amir Shee
[arXiv:2602.21134](#) (2026).
4. *Interplay of activity and non-reciprocity in tracer dynamics: From non-equilibrium fluctuation-dissipation to giant diffusion*
Subhajit Paul, Manish Patel, Debasish Chaudhuri
[arXiv:2601.03591](#) (2026).
5. *Dynamical metastability and re-entrant localization of trapped active elements with speed and orientation fluctuations*
Manish Patel, Amir Shee, Debasish Chaudhuri
Phys. Rev. Research **7**, 013126 (2025).
6. *Exact moments for trapped active particles: inertial impact on steady-state properties and re-entrance*
Manish Patel, Debasish Chaudhuri
New J. Phys. **26**, 073048 (2024).
7. *Exact moments and re-entrant transitions in the inertial dynamics of active Brownian particles*
Manish Patel, Debasish Chaudhuri
New J. Phys. **25**, 123048 (2023).

Grants & Fellowships

- Received a travel grant from the Homi Bhabha National Institute (HBNI) under the Financial Travel Assistance scheme to attend the CECAM workshop in Lausanne, Switzerland.
- Received a travel grant from the Anusandhan National Research Foundation (ANRF) under the [International Travel Support](#) scheme to attend the conference in Edinburgh, Scotland.
- Awarded with **L. K. Panda Fellowship** by Institute of Physics, Bhubaneswar, India (2021) for achieving the highest marks in the Pre-Doctoral Program.

- Qualified **JEST-2020** : National entrance test for higher education.
- Received **INSPIRE-SHE** from the Government of India (2014).

Computational & Theoretical Skills

- **Analytical methods** Langevin dynamics, Fokker–Planck equations, Green’s function based method, stochastic thermodynamics, stochastic field theory, hydrodynamics
- **Programming** C, Python
- **Parallel computing** OpenMP
- **Simulation** Molecular dynamics, agent-based simulations of active particles, Metropolis Monte Carlo
- **Scientific software** Mathematica, LAMMPS, Matplotlib, OVITO
- **Operating systems** Linux, HPC / cluster computing

Teaching Assistant

- Teaching Assistant for “Advanced Statistical Mechanics,” Institute of Physics, Bhubaneswar, January–August 2022.
- Teaching Assistant for “Advanced Statistical Mechanics,” Institute of Physics, Bhubaneswar, August–December 2023.
- Teaching Assistant for “Advanced Statistical Mechanics,” Institute of Physics, Bhubaneswar, August–December 2024.

Academic Visits & Talks

- Visited **Prof. Ignacio Pagonabarraga**’s group at University of Barcelona, Spain from Dec. 15–18, 2025 and presented a talk on “*Effect of timescales in active systems*”.
- Visited **Prof. Fernando Peruani**’s group at CY Cergy Paris University, France on Dec. 12–13, 2025 and presented a talk on “*Effect of timescales in active systems*”.
- Visited **Prof. Pierre Sens**’s group at Institut Curie, Paris, France from Dec. 8–12, 2025 and presented a talk on “*Effect of timescales in active systems*”.

Conferences & Posters

- Presented a poster on “*Role of inertia in active systems*” at **11th Indian Statistical Physics Community Meeting**, ICTS, Bangalore, India, Apr. 27–29, 2026.
- Presented a poster on “*From single particle to harmonic chain: Exploring the inertial impact in active Brownian particle systems*” at **Unveiling self-phoretic colloidal propulsion**, CECAM-HQ-EPFL, Lausanne, Switzerland, Dec. 3–5, 2025.
- Presented a poster on “*The impact of inertia on active Brownian particles*” at **Physics of Biological Systems**, University of Edinburgh, UK, May 27–29, 2025.
- Presented a poster on “*Re-entrant transitions in inertial dynamics of active Brownian particles*” at **Spring College in the Physics of Complex Systems**, ICTP, Italy, Feb. 17–Mar. 14, 2025.
- Presented a poster on “*Re-entrant transitions in inertial dynamics of active Brownian particles*” at **Frontiers in Non-Equilibrium Physics II**, IMSc, India, Jan. 7–11, 2025.
- Presented a poster on “*Re-entrant transitions in inertial dynamics of active Brownian particles*” at **Physics of Life: Active and Living Matter**, India, Feb. 8–10, 2024.

Schools & Workshops

- **Unveiling self-phoretic colloidal propulsion**, CECAM-HQ-EPFL, Lausanne, Switzerland, December 3, 2025 - December 5, 2025.
- **Spring College in the Physics of Complex Systems: SMR 4056**, ICTP, Italy, Feb. 17–Mar. 14, 2025. Passed with highest grade **E (Excellent)**.

- **Bangalore School on Statistical Physics XIV**, RRI-ICTS, India, Sep. 11–22, 2023.
- **School on Stochastic Thermodynamics in Biology**, IIT Bombay, India, Nov. 28–Dec. 2, 2022.
- **Bangalore School on Statistical Physics XIII**, ICTS, India, July 11–22, 2022 (online).
- **Workshop on Stochastic Thermodynamics III**, University of Tokyo, Japan, May 26–June 3, 2022 (online).
- **Bangalore School on Statistical Physics XII**, ICTS, India, June 28–July 9, 2021 (online).

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